



Regulating framework aluminum location towards boosted light olefins generation in ex-situ catalytic pyrolysis of low-density polyethylene

Ting Wang^{a,b,1}, Xiang Feng^{a,1}, Dong Lin^a, Yan Li^a, Jingyuan Shang^a, Jinqing Zhang^a, Shuo Li^c, Yibin Liu^a, Hui Zhao^a, Zhe Ma^a, Xiaobo Chen^a, Xingong Zhang^{b,*}, Chaohe Yang^{a,*}

^a State Key Laboratory of Heavy Oil Processing, China University of Petroleum, Qingdao 266580, China

^b Qingdao Huicheng Environmental Technology Co., Ltd., Qingdao 266580, China

^c College of Chemistry & Chemical Engineering, Yantai University, Yantai 264005, China

ARTICLE INFO

Keywords:

ZSM-5

Framework aluminum location

LDPE

Light olefins

Catalytic pyrolysis

ABSTRACT

Amidst the global focus on recycling and treatment of waste plastics for resource sustainability, it remains an enormous challenge to convert waste plastics into light olefin. Herein, the impact of framework Al (Al_F) location in ZSM-5 catalyst, controlled by regulating Al sources, on the ex-situ catalytic pyrolysis of low-density polyethylene (LDPE) has been investigated. Based on multi-techniques, such as ^{27}Al MAS NMR, ICP-AES, Co-UV-vis-DRS, XRD, SEM, N_2 adsorption-desorption, NH_3 -TPD, Py-IR and FTIR. It is found that the Al_F located at the channel intersections show more Al pairs, and this location offers ample space for secondary reactions, which include Diels-Alder reaction, cyclization reaction and dehydrogenation reaction, resulting in the low yield of light olefins (10.6 wt%) and the generation of light alkanes and aromatics from light olefins. In contrast, Al_F located in the straight and sinusoidal channels exists mainly in the form of single Al. The substantial steric hindrance for this location facilitates the cracking reaction, mainly referring to the cracking reactions of LDPE that produce light olefins and generate enhanced yield of light olefins (52.8 wt%). These results provide new perspectives towards boosted light olefins generation in ex-situ catalytic pyrolysis of LDPE.

1. Introduction

Plastics, as one of the three major synthetic materials, have found wide-ranging applications in various facets of human life [1]. In 2022, the global production of plastic products reached approximately 400 million tons. Alarmingly, over 300 million tons of plastic waste were generated worldwide, with a meager 9 % being recycled primarily through incineration, landfill, or disposal methods. The improper disposal of this substantial volume of plastic waste not only poses a severe threat to the environment but also represents a significant squandering of energy and resources [2,3].

Catalytic cracking of waste plastics into fuel and chemical raw materials has garnered significant attention as a non-oil route to chemical production in recent years, due to its advantageous characteristics, such as high H/C ratio, abundant availability, ease of access, and cost-effectiveness. Various catalysts have been employed for this purpose, including zeolites [4–7], oxides [8], clay [9,10], and transition metal-supported catalysts [11]. Oxide and clay catalysts predominantly yield

liquid hydrocarbons [12,13], such as gasoline and diesel, while transition metal-supported catalysts under hydrogenation conditions produce primarily saturated liquid products [14,15]. In contrast, highly selective gaseous hydrocarbons (mainly olefins) can be obtained utilizing ZSM-5 as catalysts, along with high octane number gasoline fractions (C_5 – C_{12}) [16–18].

In general, the performance of zeolites hinges on factors like the number, type, and strength of acid sites as well as pore structure. An equally pivotal factor is the distribution of aluminum (Al) atoms within the zeolite framework. The ZSM-5 catalyst, which features a Z-shaped 10-membered ring channel parallel to the a-axis and c-axis and a straight 10-membered ring channel parallel to the b-axis [19], exhibits greater steric hindrance within channels when compared to the intersections of channels due to the smaller pore size. Additionally, ZSM-5 belongs to the orthorhombic system and possess 12 non-equivalent T-sites within their framework. This means that Al atoms can be distributed in straight channels, sinusoidal channels, and channel intersections. According to the value of n in the $Al-O-(Si-O)_n-Al$ sequence, which represents the

* Corresponding authors.

E-mail addresses: zhangxingong@hcept.com (X. Zhang), yangch@upc.edu.cn (C. Yang).

¹ These authors contributed to this work equally.

distance between the adjacent framework aluminum, the framework aluminum is divided into paired aluminum (Al pairs) and single aluminum. When n is 1 or 2, the aluminum atoms in the sequence are called Al pairs. When the value of n is greater than or equal to 3, the aluminum atoms in the sequence are called single Al [20–22].

To date, researchers have employed various techniques, such as introducing heteroatoms, altering the structure directing agent [23–26], changing the silicon and aluminum sources [21,27], and employing post-treatment methods like dealuminization, to regulate the distribution of aluminum in ZSM-5. For instance, Sazama et al. [28] synthesized H-ZSM-5 zeolites with similar Si/Al ratios but different contents of Al pairs and single Al. They found that Al pairs favored the oligomerization and hydrogen transfer reaction of 1-butene, while single Al facilitated the cracking of 1-butene. Yokoi et al. [29] used tetrapropylammonium hydroxide (TPAOH) as a structure directing agent to synthesize H-ZSM-5 and discovered that the addition of sodium enriched the single Al primarily in straight and sinusoidal channels, whereas in the sodium-free system, the single Al was predominantly concentrated at channel intersections, which was more conducive to the cracking of 3-methylpentane and aromatic disproportionation reactions. Given that the cracking of waste plastics into low-carbon olefins primarily occurs at acidic sites derived from framework aluminum, improving the performance of ZSM-5 catalysts by precisely controlling the location of framework aluminum is a promising avenue and is also highly desired.

In this study, the structure-performance relationship between Al location and ex-situ catalytic pyrolysis of LDPE was elucidated. The regulation of Al location and acid distribution in H-ZSM-5 was achieved by utilizing different aluminum sources during hydrothermal synthesis. Multi-techniques such as ^{27}Al MAS NMR, ICP-AES, UV-vis-DRS, XRD, SEM, N_2 adsorption-desorption, NH_3 -TPD, Py-IR, FTIR were employed to distinguish acid sites in the straight and sinusoidal channels and those at the channel intersections, and the type of Al atoms in the H-ZSM-5 framework. Subsequently, a correlation between the reaction pathways in the ex-situ catalytic pyrolysis of LDPE and the distribution of acid sites among the straight/sinusoidal channels and intersections and the type of framework aluminum in H-ZSM-5 was established. Al sites located at the channel intersections offers ample space for secondary reactions, resulting in the consumption of light olefins and the generation of light alkanes and aromatics. In contrast, Al sites located in the straight and sinusoidal channels shows substantial steric hindrance for this location, facilitating the cracking reaction and generating enhanced yield of light olefins. The insights in this work is of prime scientific and industrial significance to the design of superior catalysts for the ex-situ catalytic pyrolysis of LDPE into light olefins.

2. Experimental section

2.1. Catalysts preparation

ZSM-5 zeolites (Table S1) with the similar $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratios were synthesized by altering the aluminum sources. Tetraethyl orthosilicate (TEOS, purity > 98.5 %, Sinopharm Chemical Reagent Co. Ltd) was employed as the primary Si precursor. Aluminum sources include Al (NO_3) $_3 \cdot 9\text{H}_2\text{O}$ (purity 98 %, Sinopharm Chemical Reagent Co. Ltd), AlCl_3 (purity 99 %, Sinopharm Chemical Reagent Co. Ltd), $\text{Al}(\text{OH})_3$ (containing 65 % Al_2O_3 , Sinopharm Chemical Reagent Co. Ltd), and sodium aluminate (containing 22 % SiO_2 and 22 % Na_2O , Sinopharm Chemical Reagent Co. Ltd). Tetrapropylammonium hydroxide (TPAOH, 40 % solution in water, Fluka) played the crucial role as the structure-directing agent. The synthesis processes were conducted in autoclaves at a temperature of 170 °C under autogenous pressure. Subsequently, the zeolite products underwent thorough washing with distilled water and were then subjected to drying at 80 °C. To complete the preparation, the samples were calcined at 550 °C in a stream of air for a duration of 6 h.

For the production of H-ZSM-5, the directly calcined samples underwent ion exchange twice with a NH_4NO_3 aqueous solution (1 M) at

80 °C for 4 h. Following this exchange, they were subjected to a final calcination step at 550 °C for 6 h in air.

Co-ZSM-5 were prepared by following the procedures described by Dedecek and co-workers [30,31], which were used to detect the location of Al by Co substitution. Briefly, H-ZSM-5 were treated by aqueous NaNO_3 solution (1 M) at 80 °C for 7 h, and the Na-ZSM-5 samples after ion-exchange obtained here were then ion-exchanged with aqueous Co (NO_3) $_2$ solution (0.05 M) at 80 °C for 7 h. After the ion exchanges, the Co-ZSM-5 samples were then washed in distilled water for three times and dried in air. The exchange procedure was repeated twice to obtain Co-ZSM-5.

2.2. Characterizations

The crystal structures of the zeolite samples were elucidated through XRD measurements, utilizing Cu K α radiation and a Bruker D8 Advance instrument. The chemical $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of the samples were determined by Rigaku Model 3271E X-ray fluorescence diffractometer (XRF) of Bruker. For N_2 adsorption-desorption analysis, a Micromeritics ASAP 2420 instrument was employed. The total surface area was calculated employing the Brunauer-Emmett-Teller (BET) equation, while the micropore surface area and volume were ascertained utilizing the t-plot method. To discern the pore size distribution within the zeolites, a nonlinear density functional theory (DFT) approach was applied. Scanning electron microscopy (SEM) images were captured using a Hitachi S-4800 field-emission microscope. The total acid content and the distribution of strong and weak acids within the zeolites were determined via ammonia temperature-programmed desorption (NH_3 -TPD) methodology, utilizing a Micromeritics AutoChem II 2920 chemisorption analyzer. Furthermore, the acid type associated with the zeolites was identified by analyzing infrared spectrograms of adsorbed pyridine (Py-IR) on a Thermo Fisher Scientific Nicolet 6700 Fourier-transform infrared spectrometer. Before analysis, the samples underwent a process involving the removal of adsorbed water at 300 °C for 3 h, followed by the adsorption of pyridine at room temperature for 24 h. Physically adsorbed pyridine was subsequently eliminated at 150 °C in a vacuum drying chamber. The chemical property of the samples was determined by IS50 Fourier transform infrared spectrometer (Nicolet, USA). The sample powders were dried overnight and then grinded with KBr at a mass ratio of 1: 100. The pure KBr self-supporting sheet and the mixed powder self-supporting sheet were pressed and baked under the infrared lamp for 10 min. The self-supporting sheet containing the sample to be tested was placed in the sample pool to test. The FTIR spectra of the samples were obtained after deducting the background.

For the acquisition of Magic Angle Spinning (MAS) nuclear magnetic resonance (NMR) spectra of ^{27}Al , a high-resolution Avance III 600 MHz wide bore spectrometer from Bruker was employed, operating within a magnetic field of 14.2 T, utilizing a single pulse sequence. The chemical shifts in the ^{27}Al MAS NMR spectra were referenced to an aqueous solution of $\text{Al}(\text{NO}_3)_3$. These spectra were acquired at a spinning rate of 13 kHz, with a $\pi/12$ pulse width of 1.2 μs , and a total of 8000 scans were accumulated. The ultraviolet-visible diffuse reflectance spectra (UV-vis-DRS) were collected on a UV-3600i plus ultraviolet spectrophotometer of Shimadzu. Before each measurement, the hydrated Co-ZSM-5 samples were transformed to dehydrated form under 10^{-1} Pa and 500 °C for 5 h; the dehydrated samples were then transferred into a sample cell in a glovebox. Operating at a scan speed of 10 nm/s, a step length of 1 nm, and a slim width of 5 nm, the UV-vis-DRS spectra were collected in a differential mode referenced to their parent H-ZSM-5 zeolites. In the framework of H-ZSM-5, Al atoms can exist in the forms of Al pairs and single Al. Al pairs could be balanced by hexaaquocomplexes of Co(II) cations, while single Al could not coordinate with a Co(II)(II) cation. The concentration of single Al and Al pairs can be calculated by

$$[\text{single Al}] = [\text{Al}_{\text{total}}] - 2[\text{Co}_{\text{max}}] \quad (1)$$

$$[\text{Al pairs}] = 2[\text{Co}_{\text{max}}] \quad (2)$$

where $[\text{Al}_{\text{total}}]$ and $[\text{Co}_{\text{max}}]$ are the Al content and Co content in Co-ZSM-5, respectively. The content of Al, Na and Co were determined by inductively coupled plasma-atomic emission spectroscopy (ICP-AES, Autoscan 16, TJA).

2.3. Catalytic test

The catalysts were firstly pelletized and subsequently sieved, achieving a particle size of 0.25 mm. The catalysts were then subjected to calcination under a nitrogen stream at 550 °C for a duration of 1 h. The catalytic testing process was carried out within a fixed-bed flow-through reactor, maintaining a temperature of 550 °C and atmospheric pressure. In this reactor, 1 or 2 g of the feedstocks (i.e., LDPE) were packed in a funnel which was connected with nitrogen pipeline at the top with pressure of about 0.3 ~ 0.4 MPa and was connected with reactor tube at the bottom cutting off by ball valve. When reaction furnace reached 550 °C, reaction temperature was stable, opening the ball valve allowed instant feeding. At the same time, the carrier gas switch of another nitrogen pipeline was opened which brought the reaction products to the collection system as carrier gas with 75 ml/min of flow rate by adjustment of mass flowmeter. The reaction products underwent comprehensive analysis utilizing a gas chromatograph (Agilent 7890A), equipped with a dimethyl siloxane capillary column, to elucidate their composition and characteristics. For gas products, the temperature of oven and detector were 50 °C and 190 °C, respectively. For liquid products, the temperature of oven and detector were 35 °C and 350 °C, respectively. For C₅-C₁₂ components, the temperature of oven and detector were 35 °C and 250 °C, respectively.

3. Results and discussion

3.1. Aluminum location in H-ZSM-5 zeolite

It is reported by Dedecek [28] that the relationship between local geometry of the AlO_4 tetrahedron and the NMR chemical shift are relatively intricate, and the Al location should be determined by multi-techniques. The ^{27}Al MAS NMR spectra of the synthesized H-ZSM-5

zeolite samples were analyzed to elucidate differences in Al_F distribution in various T-sites, as depicted in Fig. 1. The NMR chemical shifts of various T-sites could be theoretically calculated on the basis of average T-O-T angles. The chemical shifts of 12 T-sites in the orthorhombic H-ZSM-5, which are proximal to the actual spectra, are displayed in Fig. 1b [32]. The ^{27}Al MAS NMR spectra can be deconvoluted into five peaks at 52, 53, 54, 56 and 58 ppm [26,28,29,32,33]. The peak at 54 ppm ($\text{Al}_{(54)}$) was related to the acid sites in the channel intersections, whereas the peak at 56 ppm ($\text{Al}_{(56)}$) could be the acid sites in the straight and sinusoidal channels, as shown in Fig. 1a. For the different ZSM-5 samples, the deconvoluted peaks are shown in Fig. 1b. Based on the peak areas, the proportions of $\text{Al}_{(54)}$, $\text{Al}_{(56)}$ and the ratio of $\text{Al}_{(54)}/\text{Al}_{(56)}$ are represented in Fig. 1c.

Evidently, the fraction of the peak area at 56 ppm for H-ZSM-5 zeolite samples utilizing different aluminum sources follows this order: $\text{ZSM-5-Al}(\text{NO}_3)_3 > \text{ZSM-5-AlCl}_3 > \text{ZSM-5-Al}(\text{OH})_3 > \text{ZSM-5-NaAlO}_2$, while the fraction of the peak area at 54 ppm follows this order: $\text{ZSM-5-NaAlO}_2 > \text{ZSM-5-Al}(\text{OH})_3 > \text{ZSM-5-AlCl}_3 > \text{ZSM-5-Al}(\text{NO}_3)_3$, respectively. The results show that the ZSM-5-Al(NO_3)₃ exhibits a higher distribution of Al_F in the straight and sinusoidal channels, whereas ZSM-5-NaAlO₂ possesses more Al_F atoms situated in the channel intersections. The order of the $\text{Al}_{(54)}/\text{Al}_{(56)}$ ratio aligns with that of $\text{Al}_{(54)}$ (Fig. 1c).

According to the equation (1) and (2), the contents of various aluminum species in the Co-ZSM-5 zeolites are given in Fig. 2b, which illustrates that the majority of Al_F exists in the form of Al pairs (ca. 50–70 %). The order of proportion of single Al is as follows: $\text{ZSM-5-Al}(\text{NO}_3)_3 > \text{ZSM-5-AlCl}_3 > \text{ZSM-5-Al}(\text{OH})_3 > \text{ZSM-5-NaAlO}_2$. In other words, the ZSM-5-Al(NO_3)₃ owns highest content of single Al. The value of $(2\text{Co}+\text{Na})/\text{Al}$ is close to the fraction of Al pairs (i.e., $2\text{Co}/\text{Al}$), suggesting that the sodium content in Co-ZSM-5 is negligible and the Al pairs are mainly balanced by Co(II) cations. It is then possible to estimate the distribution of Al_F among various framework T-sites through identifying the state of Co(II) by UV-vis-DRS spectra because Al pairs are coordinated with Co(II) cations.

The UV-vis-DRS spectra of Co(II) cations in Co-ZSM-5 are deconvoluted and shown in Fig. 2c [31,34]. Seven or eight bands are obtained, which can be sorted into three types of Co(II) ions coordinated with different Al T-sites [21,26,31,35,36]: α -type Co(II) ions in the straight channel are linked with a single band at 15100 cm^{-1} ; β -type Co(II) ions

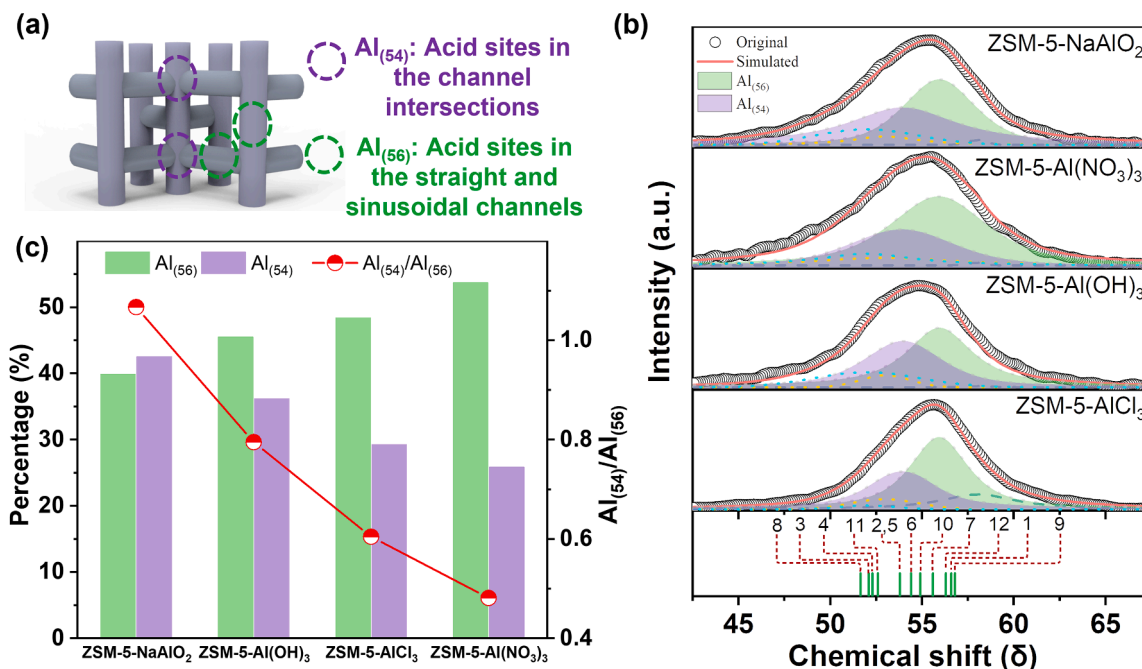


Fig. 1. (a) Diagram of different aluminum sites, (b) ^{27}Al MAS NMR spectra, (c) distribution of $\text{Al}_{(54)}$ and $\text{Al}_{(56)}$ of the H-ZSM-5 zeolite samples.

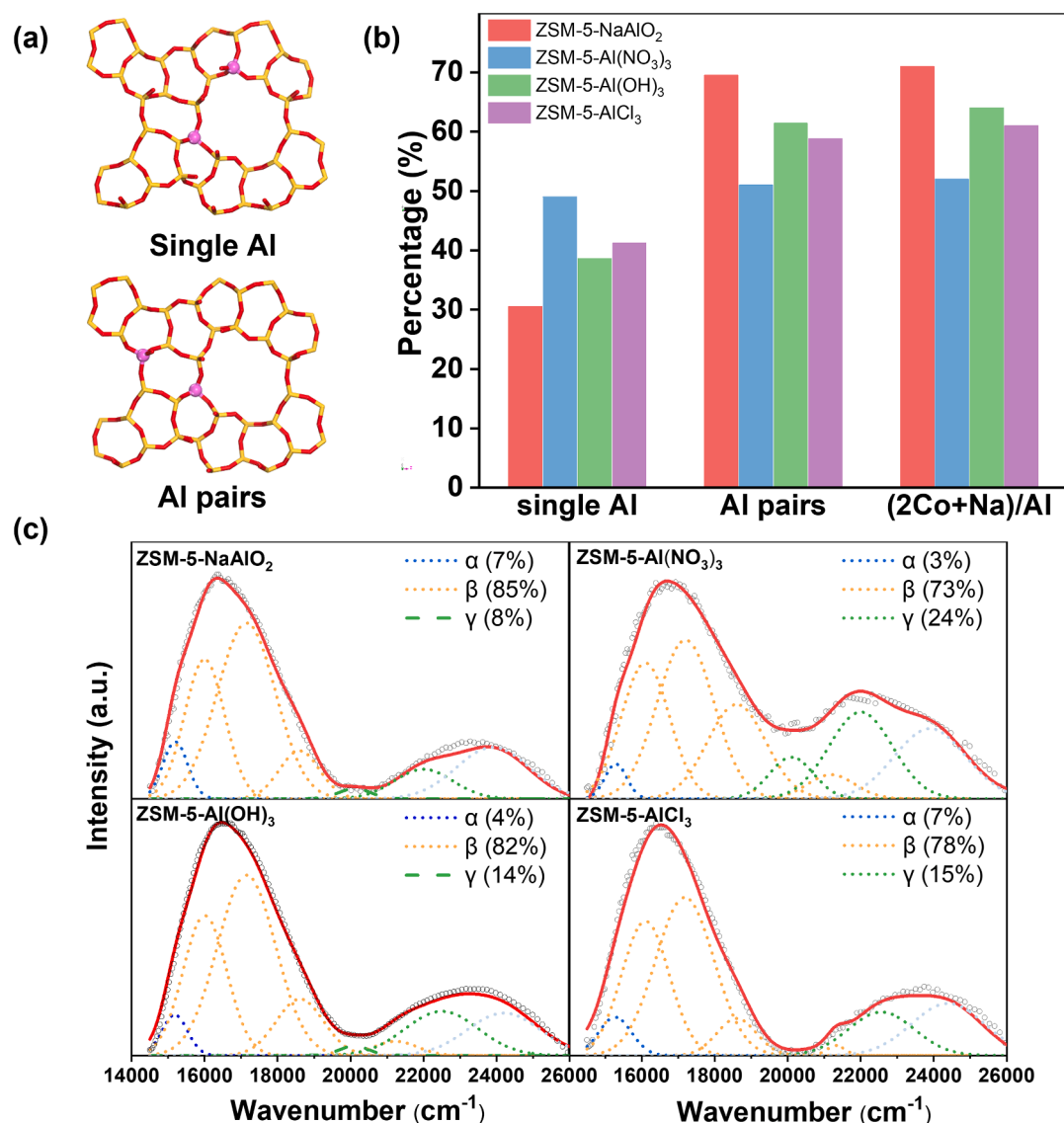


Fig. 2. (a) Schematic arrangement of Al atoms in Al pairs (Al pairs), and single Al atoms, (b) distribution of single Al and Al pairs of the H-ZSM-5 zeolites, (c) UV-vis-DRS spectra of Co(II) ions of Co-ZSM-5 zeolites.

in the channel intersections are characterized by four bands at 16000, 17150, 18600, and 21200 cm⁻¹; and γ -type Co(II) ions in the sinusoidal channels are related to the bands at 20100 and 22000 cm⁻¹. Fig. 2c illustrates that the difference in the distribution of Al_F among various framework T-sites of four zeolite samples is relatively obvious, though in all cases a majority of Al_F may be located at channel intersections. ZSM-5-Al(NO₃)₃ possesses the largest number of Al_F locating in the straight and sinusoidal channels (27%), while ZSM-5-NaAlO₂ has most Al_F locating in the channel intersections.

For the ZSM-5 zeolite samples, a correlation of the Al₍₅₄₎/Al₍₅₆₎ ratio estimated from ²⁷Al MAS NMR spectra with the $\beta/(\alpha+\gamma)$ ratio from Co-UV-vis-DRS is illustrated in Fig. 3. In general, the ratio of Al₍₅₄₎/Al₍₅₆₎ is directly proportional to the value of $\beta/(\alpha+\gamma)$. Because of the multiple controlling factors during zeolite synthesis and the complexity of characterizing Al location, it is difficult to accurately assign Al_F to specified T-sites in ZSM-5 at present. However, the linear correlation illustrated in Fig. 3 does suggest that both ²⁷Al MAS NMR and Co-UV-vis-DRS spectra can provide valuable and consistent information for the location of Al_F in ZSM-5. Consequently, we can infer that the ZSM-5-NaAlO₂ possesses the highest content of Al pairs in the channel intersections, whereas the ZSM-5-Al(NO₃)₃ contains the highest content of Al pairs in the straight

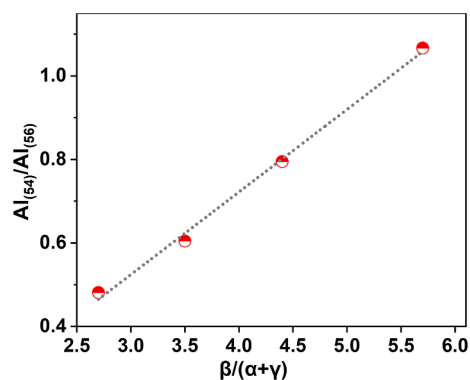


Fig. 3. Relationship between the $\beta/(\alpha+\gamma)$ and Al₍₅₄₎/Al₍₅₆₎ ratio.

and sinusoidal channels.

The above analysis results of ²⁷Al MAS NMR and Co-UV-vis-DRS are credible which are in agree with findings from literatures [26,33,36]. The analysis presented here underscores the ability to tailor the

distribution of Al atoms by selecting different aluminum sources. Choosing $\text{Al}(\text{NO}_3)_3$ as aluminum source can facilitate Al_F locating in the straight and sinusoidal channels while reducing the number of Al pairs in the channel intersections.

3.2. Physico-chemical properties of H-ZSM-5 zeolite with different Al locations

As depicted in Fig. 4a, all the synthesized H-ZSM-5 zeolite catalysts exhibit a typical MFI topology, characterized by high crystallinity. Taking ZSM-5- AlCl_3 as a reference, all catalysts consistently demonstrate a relative crystallinity ranging from 90 % to 100 %, as detailed in Table S2. The sharp peak at 24.4° is evident for the samples, indicating the typical orthorhombic phase [37]. Moreover, the SEM images in Fig. 4b reveal that all the synthesized zeolite samples share a similar layered ellipsoid morphology, with crystal sizes falling within the range of 2–5 μm . The particle size distribution for zeolite sample spans less than 2 μm , and the disparity in the average particle size among samples employing various aluminum sources is less than 1 μm . As shown in Fig. 4c and Table S2, N_2 physisorption measurements elucidate that the synthesized H-ZSM-5 zeolite samples, originating from various aluminum sources, manifest analogous adsorption isotherm types. Additionally, these samples display comparable surface areas and pore volumes, along with similar $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratios within the range of 35–40, as determined by XRF analysis. Based on the above analysis, the

synthesized H-ZSM-5 zeolite samples exhibit uniform characteristics in terms of morphology, crystal size, textural properties, and $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratios. Consequently, differences in the catalytic performance of H-ZSM-5 zeolites in the ex-situ catalytic pyrolysis of LDPE to yield light olefins are likely unrelated to their morphology and textural properties.

As depicted in Fig. 4d and Table S3, NH_3 -TPD results demonstrate that the synthesized H-ZSM-5 zeolite samples possess nearly identical total acid site counts. However, the abundance of weak acid sites on the samples varies from highest to lowest as follows: ZSM-5- $\text{Al}(\text{NO}_3)_3 > \text{ZSM-5-AlCl}_3 > \text{ZSM-5-Al}(\text{OH})_3 > \text{ZSM-5-NaAlO}_2$. This order is consistent with the order of the ratio of B/L calculated by Py-IR as shown in Table S3. ZSM-5- $\text{Al}(\text{NO}_3)_3$ exhibits a greater quantity of weak acid sites compared to the other samples. The Brønsted acid sites (BAS) are primarily attributed to framework aluminum (Al_F) [38]. ZSM-5- $\text{Al}(\text{NO}_3)_3$ displays a higher BAS content compared with other three samples originating from different aluminum sources. This discrepancy may be attributed to the fact that the other three samples except for $\text{Al}(\text{NO}_3)_3$ contain a higher proportion of Al_L species that are unable to form BAS.

The chemical bonds in the synthesized H-ZSM-5 zeolite samples were identified according to the FTIR spectra in the low wavenumber region, as shown in Fig. 4e. It can be found that all zeolite samples have strong infrared absorption bands at 450, 550, 796, 1080 and 1220 cm^{-1} . The band at 450 cm^{-1} is attributed to the bending vibration inside the T (representing the tetrahedral central atom) tetrahedron [39]. The peak at 550 cm^{-1} is the characteristic band of the structure of MFI type

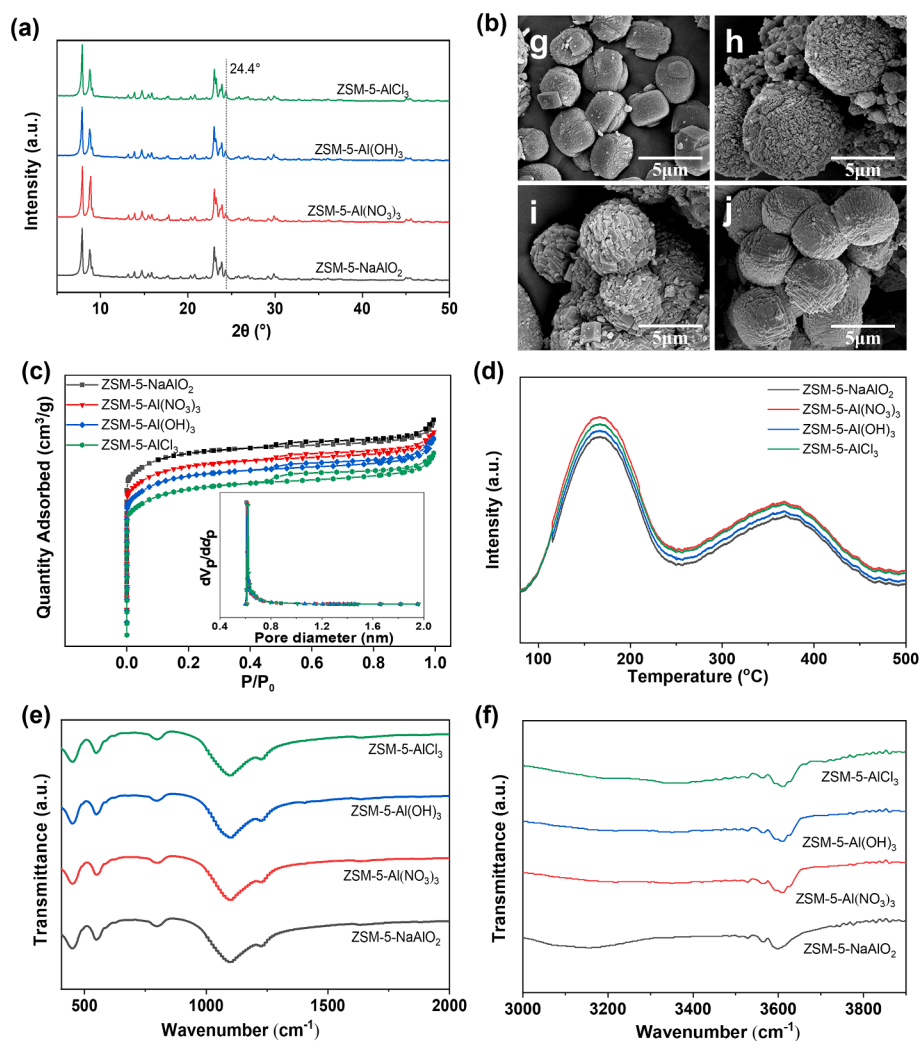


Fig. 4. (a) XRD patterns of the synthesized H-ZSM-5 zeolite samples, (b) SEM images of (g) ZSM-5- NaAlO_2 , (h) ZSM-5- $\text{Al}(\text{NO}_3)_3$, (i) ZSM-5- $\text{Al}(\text{OH})_3$, (j) ZSM-5- AlCl_3 , (c) N_2 adsorption–desorption isotherms, (d) NH_3 -TPD profiles, infrared spectra in the range of (e) 400–2000 cm^{-1} and (f) 3000–3900 cm^{-1} .

zeolite, which is attributed to the external asymmetric stretching vibration of the double five-membered ring structure [40,41]. The peak at 796 cm^{-1} is attributed to the external symmetric stretching vibration of the T tetrahedron. The peaks at 1080 cm^{-1} and 1220 cm^{-1} are attributed to the internal and external asymmetric stretching vibration of the Si-O-T bond, respectively [42]. By comprehensive analysis of the spectra in this interval, all samples possess crystal structure of typical MFI type zeolite.

From the infrared spectra in the range of $3000\text{--}3900\text{ cm}^{-1}$ (Fig. 4f), it can be observed that all zeolite samples have band at 3610 cm^{-1} , which belongs to the vibration spectrum band of Brønsted acid generated by Si-OH-Al bristle hydroxyl group. The peak intensity of ZSM-5-Al(NO_3)₃ is larger than others, and that of ZSM-5-NaAlO₂ is the smallest, which is consistent with the characterization results of Py-IR in Table S3.

3.3. The effect of aluminum location in ex-situ catalytic pyrolysis of LDPE to light olefins

Given the significant influence of aluminum location on product distribution, Fig. 5a illustrates the catalytic performance of ex-situ catalytic pyrolysis of LDPE. For all samples, the conversion of LDPE reaches 100 %, yielding C₂–C₄ light olefins, C₁–C₄ light alkanes, aromatics C₅–C₁₂, non-aromatics C₅–C₁₂, $\geq\text{C}_{13}$ components together with coke. The yield of C₂–C₄ light olefins is as follows: ZSM-5-Al(NO_3)₃ (40.2 wt%) > ZSM-5-AlCl₃ (36.7 wt%) > ZSM-5-Al(OH)₃ (35.1 wt%) > ZSM-5-NaAlO₂ (29.1 wt%). Conversely, the yield of C₁–C₄ light alkanes follows the reverse order: ZSM-5-Al(NO_3)₃ (30.6 wt%) < ZSM-5-AlCl₃ (32.4 wt%) < ZSM-5-Al(OH)₃ (33.9 wt%) < ZSM-5-NaAlO₂ (39.2 wt%). The order of the yield of aromatics C₅–C₁₂ is the same with C₁–C₄ light alkanes: ZSM-5-Al(NO_3)₃ (22.3 wt%) < ZSM-5-AlCl₃ (24.5 wt%) < ZSM-5-Al(OH)₃ (25.1 wt%) < ZSM-5-NaAlO₂ (26.1 wt%). In order to enhance the reliability of data, the average yield of light olefins and standard deviation of repeated experiment were shown in Fig. S1. The variation trend of

light olefins yield of repeated experiments is the same with above data. Also, the standard deviation of repeated experiments is less than 2 % which prove the reliability of data.

Compared with other samples, ZSM-5-Al(NO_3)₃ possesses a greater number of acid sites situated in the straight and sinusoidal channels, and a lower content of Al pairs. This configuration not only synergistically promotes the production of light olefins but also introduces higher steric hindrance in the straight and sinusoidal channels. This inhibits secondary reactions such as Diels-Alder reaction, cyclization reaction and dehydrogenation reaction, which consume light olefins and generate light alkanes and aromatics [43–45]. Conversely, ZSM-5-NaAlO₂ exhibits the highest proportion of Al located in the channel intersections and highest content of Al pairs which mainly exists in the channel intersections. This unique location favors the above secondary reactions that result in light alkanes and aromatics.

Fig. 5b also reveals that propylene has highest proportion among light olefins, followed by ethylene, while butylene accounts for the smallest fraction. The sequence of propylene and butylene yields is as follows: ZSM-5-Al(NO_3)₃ > ZSM-5-AlCl₃ > ZSM-5-Al(OH)₃ > ZSM-5-NaAlO₂. Ethylene yields remain nearly consistent among the four samples. For propylene and butylene, they are primarily generated through the β -scission of bimolecular pathway [46], and thus are highly dependent on aluminum location. In the ZSM-5-Al(NO_3)₃, the presence of more acid sites in the straight and sinusoidal channels, coupled with fewer Al pairs, impedes secondary reactions such as Diels-Alder reaction which consumes propylene and butylene. Consequently, ZSM-5-Al(NO_3)₃ catalyst yields more propylene and butylene and less aromatics. The ethylene is mainly produced by monomolecular pathway [47] and thermal cracking which are less affected by aluminum location resulting in the similar yield.

The light olefins are the main products of β -scission, while the aromatics are the main products of secondary reactions. The ratio of light olefins/aromatics could confirm the reaction proportions on different

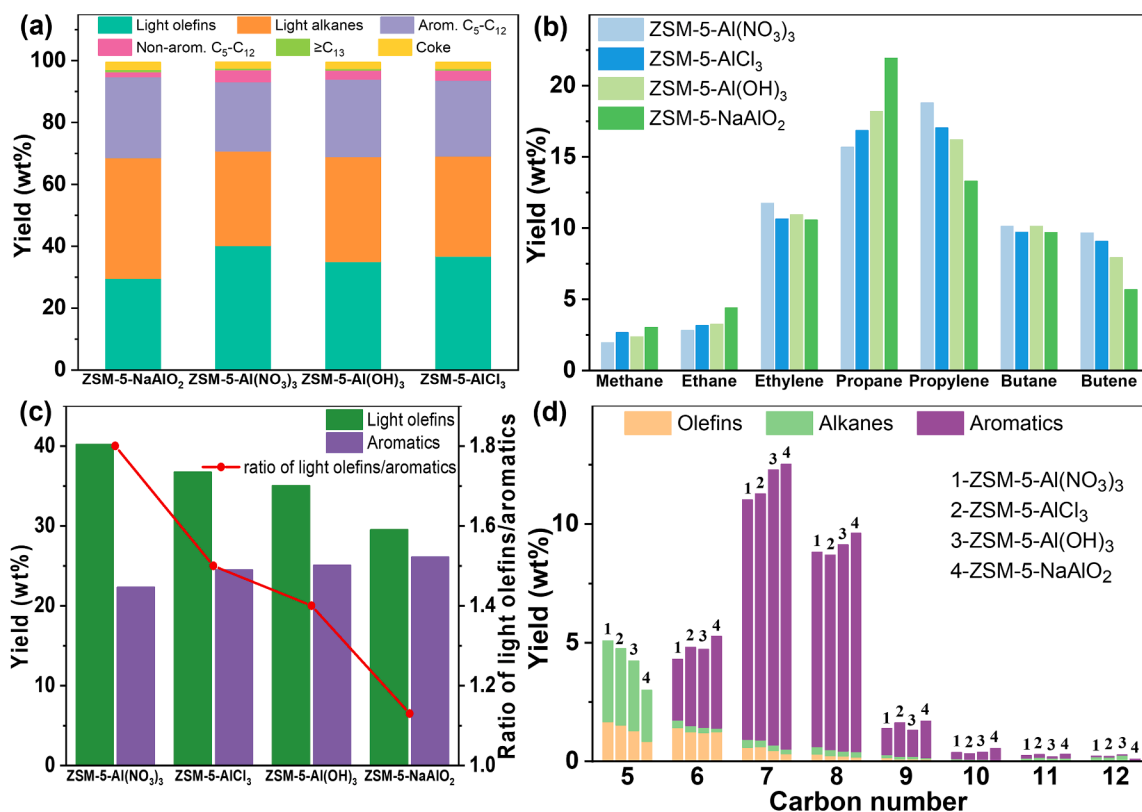


Fig. 5. (a) Production distribution in ex-situ catalytic pyrolysis of LDPE over H-ZSM-5 zeolite samples, (b) distribution of light alkanes and olefins, (c) the yield of light olefins and aromatics, and the ratio of light olefins/aromatics, and (d) distribution of monomer hydrocarbon in the range of C₅–C₁₂.

framework aluminum location [48], as depicted in Fig. 5c. The ratio of light olefins/aromatics of ZSM-5- $\text{Al}(\text{NO}_3)_3$ is highest, which illustrates that the proper aluminum location can promote chain scission reaction and inhibit the secondary reactions.

Fig. 5d displays the monomer hydrocarbon distribution of C_5 - C_{12} components. Notably, aromatics particularly within the C_6 - C_8 range have the highest yield, exemplified by benzene, methylbenzene, and p-xylene. For example, propylene and butadiene are converted to methylcyclohexene by Diels-Alder reaction, then form methylbenzene by dehydrogenation reaction [44,45]. Few butadiene is found with the yield of less than 0.1 wt% which proves that above reactions occur with high probability. Olefins primarily distributed within the C_5 - C_7 range have the second-highest yield, and alkanes yield the least. This is attributed to the fact that most of C_8^+ olefins and alkanes crack into light olefins and light alkanes through β -scission [43]. Among these reactions, Diels-Alder reaction and cyclization reaction require less steric hindrance [49,50]. Hence, in the ZSM-5- $\text{Al}(\text{NO}_3)_3$, the distribution of Al_F in straight/sinusoidal channels with larger steric hindrance does not support extensive reactions which generate aromatics.

Fig. 6 illustrates the impact of different Al (i.e., straight/sinusoidal channels and channel intersections) proportions on the yields of primary products in the ex-situ catalytic pyrolysis of LDPE. Higher yields of propylene and butylene are observed when there is a higher proportion of $\text{Al}_{(56)}$ and single Al in straight and sinusoidal channels, while the ethylene yield remains rather similar. The cumulative yields of C_2 - C_4 light olefins surpass 40 wt% of the total gas product, reaching up to 60 wt%, which is promising to increase the supply of light olefins from a technical perspective.

3.4. Discussion on ex-situ catalytic pyrolysis mechanism of LDPE to light olefins and application prospect

From the above analysis, the mechanism of ex-situ catalytic pyrolysis of LDPE is briefly discussed, as shown in Fig. 7. In the inert atmosphere of 550 °C, LDPE first pyrolyzed into intermediate products (mixture of long-chain alkanes and long-chain alkenes, small amount of cyclic hydrocarbons) [51,52]. Firstly, the long-chain molecules form carbenium ions by obtaining hydrogen proton from acid sites. Then carbenium ions undergo β -scission reactions to produce primary products mainly light olefins and light alkanes [43–45]. The as-produced light olefins then occurs the secondary reactions, especially on the Al pairs in the channel intersections. Through Diels-Alder reaction, cyclization reaction and dehydrogenation reaction, light olefins transform into aromatics and light alkanes, resulting the reduction of light olefins and the increase of aromatics and light alkanes [43–45]. This step relies on less steric hindrance and the existence of more Al pairs in the channel intersections. By controlling the framework Al location, it is possible to regulate reaction pathway. Furthermore, the catalytic performance of ZSM-5- $\text{Al}(\text{NO}_3)_3$ catalyst can be further boosted by optimization (Fig. 7b-c). It is

noted that the C_2 - C_4 light olefins yield can be further improved by reducing the catalyst weight to 1 g, resulting the higher C_2 - C_4 light olefins yield of 52.8 %. Compared with the ZSM-5- NaAlO_2 and ZSM-5- $\text{Al}(\text{NO}_3)_3$ -2g, Al_F of ZSM-5- $\text{Al}(\text{NO}_3)_3$ -1g declines in the number and mainly locates in the straight and sinusoidal channels in which single Al take a large proportion. Lower catalyst loading not only provides enough catalytic activity for the pyrolysis of LDPE, but also restrains the happening of secondary reactions which consuming light olefins in many ways. This is mostly because the proper framework aluminum location of ZSM-5- $\text{Al}(\text{NO}_3)_3$ -1g. Notably, the C_2 - C_4 light olefins with high yields could serve as essential platform chemicals in the production of polymers. For instance, ethylene, typically produced through steam pyrolysis, forms the cornerstone of the modern chemical industry. Similarly, propylene, usually derived from naphtha cracking and propane dehydrogenation, represents a crucial foundational chemical raw material. C_4 olefins, primarily isobutene and butadiene, constitute the primary raw materials for exceptionally valuable polymers.

4. Conclusion

In this study, the influence of framework aluminum location in ZSM-5, which can be controlled by varying the aluminum sources, on the ex-situ catalytic pyrolysis of LDPE was investigated. When framework aluminum is situated in the straight and sinusoidal channels and exists mainly in the form of single Al, there is significant steric hindrance and fewer acid sites, which promotes catalytic cracking reactions. As a result, it facilitates the cracking of LDPE into light olefins and light alkanes. Conversely, when framework aluminum is positioned at the channel intersections which accommodate more Al pairs, it provides ample space and acid sites for secondary reactions. These reactions encompass Diels-Alder reaction, cyclization reaction and dehydrogenation reaction, all of which lead to the consumption of light olefins and the formation of light alkanes and aromatics. These insights shed new light on the significance of framework aluminum location within H-ZSM-5 and open up perspectives for designing enhanced catalysts for the ex-situ catalytic pyrolysis of LDPE.

CRediT authorship contribution statement

Ting Wang: Data curation, Formal analysis, Writing – original draft. **Xiang Feng:** Conceptualization, Data curation, Formal analysis, Investigation. **Dong Lin:** Formal analysis, Data curation. **Yan Li:** Data curation. **Jingyuan Shang:** Data curation. **Jinqing Zhang:** Data curation. **Shuo Li:** Data curation. **Yibin Liu:** Formal analysis, Data curation. **Hui Zhao:** Formal analysis, Data curation. **Zhe Ma:** Data curation, Formal analysis. **Xiaobo Chen:** Formal analysis. **Xingong Zhang:** Supervision, Validation. **Chaohe Yang:** Formal analysis, Funding acquisition, Investigation, Writing – review & editing.

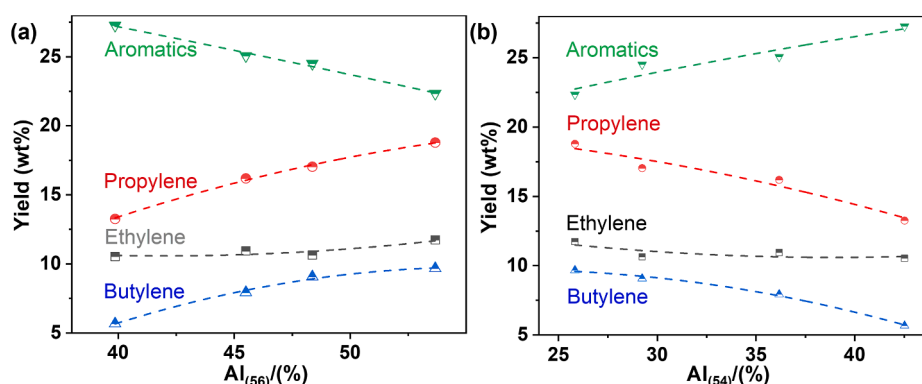


Fig. 6. Yield of ethylene, propylene, butylene, and aromatics in ex-situ catalytic pyrolysis of LDPE over H-ZSM-5 as a function of (a) $\text{Al}_{(56)}$, (b) $\text{Al}_{(54)}$.

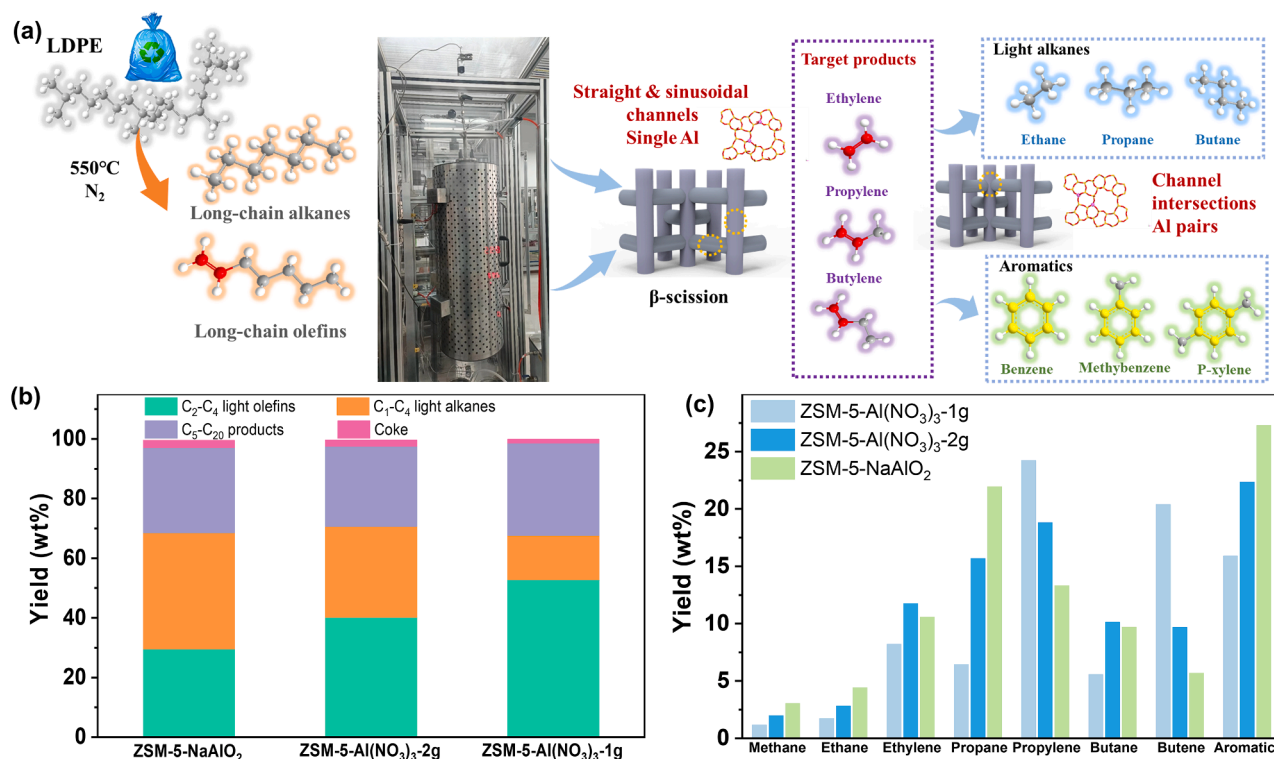


Fig. 7. (a) Plausible reaction scheme of LDPE over the synthesized H-ZSM-5 catalyst with different Al locations, (b) production distribution in ex-situ catalytic pyrolysis of LDPE over H-ZSM-5 zeolite samples, distribution of (c) light alkanes and olefins.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China-Outstanding Youth Foundation (22322814), National Natural Science Foundation of China (21978325), the Natural Science Foundation of Shandong Province (ZR2020KB006) and the Outstanding Youth Fund of Shandong Provincial Natural Science Foundation (ZR2020YQ17).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.149737>.

References

- [1] A. Rafey, K. Pal, A. Bohre, A. Modak, K. Pant, A state-of-the-art review on the technological advancements for the sustainable management of plastic waste in consort with the generation of energy and value-added chemicals, *Catalysts* 13 (2023) 24.
- [2] Q. Hou, M. Zhen, H. Qian, Y. Nie, X. Bai, T. Xia, M. Rehman, Q. Li, M. Ju, Upcycling and catalytic degradation of plastic wastes, *Cell Rep. Phys. Sci.* 2 (2021) 30.
- [3] L. Korley, T. Epps, B. Helms, A. Ryan, Toward polymer upcycling-adding value and tackling circularity, *Science* 373 (2021) 66–69.
- [4] B. Santos, D. Almeida, M. Marques, C. Henriques, Degradation of Polypropylene and Polyethylene Wastes Over HZSM-5 and USY Zeolites, *Catal. Lett.* 149 (2019) 798–812.
- [5] L. Dai, N. Zhou, Y. Lv, K. Cobb, Y. Cheng, Y. Wang, Y. Liu, P. Chen, R. Zou, H. Lei, Pyrolysis-catalysis for waste polyolefin conversion into low aromatic naphtha, *Energy Conv. Manag.* 245 (2021) 13.
- [6] L. Furda, D. Smalchenko, O. Lebedeva, Catalytic Conversion of Polyolefins in the Presence of MCM-41 Type Mesoporous Materials, *Pet. Chem.* 62 (2022) 310–315.
- [7] K. Sivagami, K. Kumar, P. Tamizhdurai, D. Govindarajan, M. Kumar, I. Nambi, Conversion of plastic waste into fuel oil using zeolite catalysts in a bench-scale pyrolysis reactor, *RSC Adv.* 12 (2022) 7612–7620.
- [8] H. Liu, H. Chen, Z. Zhang, D. Fan, Fly Ash Catalytic Pyrolysis of Waste Polypropylene Plastics, *Environ. Prot. Chem. Ind.* 30 (2010) 336–339.
- [9] A. Lopez-Uribebarrenechea, I. de Marco, B. Caballero, M. Laresgoiti, A. Adrados, A. Aranzabal, Catalytic pyrolysis of plastic wastes with two different types of catalysts: ZSM-5 zeolite and Red Mud, *Appl. Catal. B: Environ.* 104 (2011) 211–219.
- [10] D. Luiz, C. Saron, Catalytic effect of natural clays on properties and chemical structure of recycled polyethylene, *J. Mater. Cycles Waste* 24 (2022) 2545–2554.
- [11] Y. Zhang, D. Duan, H. Lei, E. Villota, R. Ruan, Jet fuel production from waste plastics via catalytic pyrolysis with activated carbons, *Appl. Energy* 251 (2019) 17.
- [12] J. Aguado, J. Sotelo, D. Serrano, J. Calles, J. Escola, Catalytic conversion of polyolefins into liquid fuels over MCM-41: Comparison with ZSM-5 and amorphous Si(O)₂-Al₂O₃, *Energy Fuels* 11 (1997) 1225–1231.
- [13] D. Serrano, J. Aguado, J. Escola, Catalytic conversion of polystyrene over HMCM-41, HZSM-5 and amorphous SiO₂-Al₂O₃: comparison with thermal cracking, *Appl. Catal. B: Environ.* 25 (2000) 181–189.
- [14] S. Kumar, A. Panda, R. Singh, A review on tertiary recycling of high-density polyethylene to fuel, *Resour. Conserv. Recycl.* 55 (2011) 893–910.
- [15] A. Kane, A. Esper, K. Searles, C. Ehm, A. Veige, Probing beta-alkyl elimination and selectivity in polyolefin hydrogenolysis through DFT, *Catal. Sci. Technol.* 11 (2021) 6155–6162.
- [16] G. Elordi, M. Olazar, P. Castano, J. Bilbao, Effect of the acidity of the HZSM-5 zeolite catalyst on the cracking of high density polyethylene in a conical spouted bed reactor, *Appl. Catal. A: Gen.* 415 (2012) 89–95.
- [17] M. Artetxe, G. Lopez, M. Amutio, G. Elordi, J. Bilbao, M. Olazar, Cracking of High Density Polyethylene Pyrolysis Waxes on HZSM-5 Catalysts of Different Acidity, *Ind. Eng. Chem. Res.* 52 (2013) 10637–10645.
- [18] J. Duan, W. Chen, C. Wang, L. Wang, Z. Liu, X. Yi, W. Fang, H. Wang, H. Wei, S. Xu, Coking-Resistant Polyethylene Upcycling Modulated by Zeolite Micropore Diffusion, *J. Am. Chem. Soc.* 144 (2022) 14269–14277.
- [19] G. Kokotailo, S. Lawton, D. Olson, D. Olson, W. Meier, STRUCTURE OF SYNTHETIC ZEOLITE ZSM-5, *Nature* 272 (1978) 437–438.
- [20] O. Han, C. Kim, S. Hong, Direct evidence for the nonrandom nature of Al substitution in zeolite ZSM-5: An investigation by ²⁷Al MAS and MQ MAS NMR, *Angew. Chem.-Int. Edit.* 41 (2002) 469–472.
- [21] J. Dedecek, V. Balgova, V. Pashkova, P. Klein, B. Wichterlova, Synthesis of ZSM-5 Zeolites with Defined Distribution of Al Atoms in the Framework and Multinuclear

- MAS NMR Analysis of the Control of Al Distribution, *Chem. Mat.* 24 (2012) 3231–3239.
- [22] J. Dedecek, Z. Sobalik, B. Wichterlova, Siting and Distribution of Framework Aluminium Atoms in Silicon-Rich Zeolites and Impact on Catalysis, *Catal. Rev.-Sci. Eng.* 54 (2012) 135–223.
- [23] Q. Zhu, J. Kondo, T. Yokoi, T. Setoyama, M. Yamaguchi, T. Takewaki, K. Domen, T. Tatsumi, The influence of acidities of boron- and aluminium-containing MFI zeolites on co-reaction of methanol and ethene, *Phys. Chem. Chem. Phys.* 13 (2011) 14598–14605.
- [24] M. Liu, T. Yokoi, M. Yoshioka, H. Imai, J. Kondo, T. Tatsumi, Differences in Al distribution and acidic properties between RTH-type zeolites synthesized with OSDAs and without OSDAs, *Phys. Chem. Chem. Phys.* 16 (2014) 4155–4164.
- [25] J. Di Iorio, R. Gounder, Controlling the Isolation and Pairing of Aluminum in Chabazite Zeolites Using Mixtures of Organic and Inorganic Structure-Directing Agents, *Chem. Mat.* 28 (2016) 2236–2247.
- [26] T. Liang, J. Chen, Z. Qin, J. Li, P. Wang, S. Wang, G. Wang, M. Dong, W. Fan, J. Wang, Conversion of Methanol to Olefins over H-ZSM-5 Zeolite: Reaction Pathway Is Related to the Framework Aluminum Siting, *ACS Catal.* 6 (2016) 7311–7325.
- [27] S. Abbasizadeh, R. Karimzadeh, Effect of Next-Nearest-Neighbor Aluminum Atoms in the HZSM-5 Framework Synthesized with Various Aluminum Sources on Liquefied Petroleum Gas Transformation to Light Olefins, *Ind. Eng. Chem. Res.* 57 (2018) 7783–7794.
- [28] S. Sklenak, J. Dedecek, C. Li, B. Wichterlova, V. Gabova, M. Sierka, J. Sauer, Aluminium siting in the ZSM-5 framework by combination of high resolution ^{27}Al NMR and DFT/MM calculations, *Phys. Chem. Chem. Phys.* 11 (2009) 1237–1247.
- [29] T. Yokoi, H. Mochizuki, S. Namba, J. Kondo, T. Tatsumi, Control of the Al Distribution in the Framework of ZSM-5 Zeolite and Its Evaluation by Solid-State NMR Technique and Catalytic Properties, *J. Phys. Chem. C* 119 (2015) 15303–15315.
- [30] J. Dedecek, D. Kaucky, B. Wichterlova, Co^{2+} ion siting in pentasil-containing zeolites, part 3. Co^{2+} ion sites and their occupation in ZSM-5: a VIS diffuse reflectance spectroscopy study, *Microporous Mesoporous Mat.* 35–6 (2000) 483–494.
- [31] J. Dedecek, D. Kaucky, B. Wichterlova, O. Gonsiorova, Co^{2+} ions as probes of Al distribution in the framework of zeolites. ZSM-5 study, *Phys. Chem. Chem. Phys.* 4 (2002) 5406–5413.
- [32] E. Lippmaa, A. Samoson, M. Magi, High-resolution ^{27}Al NMR of aluminosilicates, *J. Am. Chem. Soc.* 108 (1986) 1730–1735.
- [33] T. Biliget, Y. Wang, T. Nishitoba, R. Otomo, S. Park, H. Mochizuki, J. Rondo, T. Tatsumi, T. Yokoi, Al distribution and catalytic performance of ZSM-5 zeolites synthesized with various alcohols, *J. Catal.* 353 (2017) 1–10.
- [34] J. Dedecek, B. Wichterlova, Co^{2+} ion siting in pentasil-containing zeolites. I. Co^{2+} ion sites and their occupation in mordenite. A Vis-NIR diffuse reflectance spectroscopy study, *J. Phys. Chem. B* 103 (1999) 1462–1476.
- [35] S. Wang, P. Wang, Z. Qin, Y. Chen, M. Dong, J. Li, K. Zhang, P. Liu, J. Wang, W. Fang, Relation of Catalytic Performance to the Aluminum Siting of Acidic Zeolites in the Conversion of Methanol to Olefins, Viewed via a Comparison between ZSM-5 and ZSM-11, *ACS Catal.* 8 (2018) 5485–5505.
- [36] A. Janda, A. Bell, Effects of Si/Al Ratio on the Distribution of Framework Al and on the Rates of Alkane Monomolecular Cracking and Dehydrogenation in H-MFI, *J. Am. Chem. Soc.* 135 (2013) 19193–19207.
- [37] Y. Feng, One- and two-dimensional high-resolution solid-state NMR investigation of zeolite structures, 1991.
- [38] L. Ong, M. Domok, R. Olindo, A. van Veen, J. Lercher, Dealumination of HZSM-5 via steam-treatment, *Microporous Mesoporous Mat.* 164 (2012) 9–20.
- [39] T. Xue, Y. Wang, M. He, Facile synthesis of nano-sized $\text{NH}_4\text{-ZSM-5}$ zeolites, *Microporous Mesoporous Mat.* 156 (2012) 29–35.
- [40] D. Serrano, J. Aguado, J. Escola, A. Peral, G. Morales, E. Abella, Synthesis of hierarchical ZSM-5 by silanization and alkoxylation of protozeolitic units, *Catal. Today* 168 (2011) 86–95.
- [41] J. Zheng, Q. Zeng, Y. Yi, Y. Wang, J. Ma, B. Qin, X. Zhang, W. Sun, R. Li, The hierarchical effects of zeolite composites in catalysis, *Catal. Today* 168 (2011) 124–132.
- [42] J. Jansen, F. Vandergaag, H. Vanbekkum, Identification of ZSM-type and other 5-ring containing zeolites by ir spectroscopy, *Zeolites* 4 (1984) 369–372.
- [43] M. Abbas-Abadi, Y. Ureel, A. Eschenbacher, F. Vermeire, R. Varghese, J. Oenema, G. Stefanidis, K. Van Geem, Challenges and opportunities of light olefin production via thermal and catalytic pyrolysis of end-of-life polyolefins: Towards full recyclability, *Prog. Energy Combust. Sci.* 96 (2023).
- [44] L. Yao, J. Zhu, S. Li, Y. Ma, C. Yue, Analysis of liquid products and mechanism of thermal/catalytic pyrolysis of HDPE, *J. Therm. Anal. Calorim.* 147 (2022) 14257–14266.
- [45] Z. Dong, W. Chen, K. Xu, Y. Liu, J. Wu, F. Zhang, Understanding the Structure-Activity Relationships in Catalytic Conversion of Polyolefin Plastics by Zeolite-Based Catalysts: A Critical Review, *ACS Catal.* 12 (2022) 14882–14901.
- [46] A. Eschenbacher, F. Goodarzi, R. Varghese, K. Enemark-Rasmussen, S. Kegnæs, M. Abbas-Abadi, K. Van Geem, Boron-modified mesoporous ZSM-5 for the conversion of pyrolysis vapors from LDPE and mixed polyolefins: maximizing the $\text{C}_2\text{-C}_4$ olefin yield with minimal carbon footprint, *ACS Sustainable Chem. Eng.* 9 (2021) 14618–14630.
- [47] A. Eschenbacher, R. Varghese, E. Delikonstantis, O. Mynko, F. Goodarzi, K. Enemark-Rasmussen, J. Oenema, M. Abbas-Abadi, G. Stefanidis, K. Van Geem, Highly selective conversion of mixed polyolefins to valuable base chemicals using phosphorus-modified and steam-treated mesoporous HZSM-5 zeolite with minimal carbon footprint, *Appl. Catal. B: Environ.* 309 (2022) 121251.
- [48] Y. Xue, P. Johnston, X. Bai, Effect of catalyst contact mode and gas atmosphere during catalytic pyrolysis of waste plastics, *Energy Conv. Manag.* 142 (2017) 441–451.
- [49] M. Abbas-Abadi, M. Haghighi, H. Yeganeh, Evaluation of pyrolysis product of virgin high density polyethylene degradation using different process parameters in a stirred reactor, *Fuel Process. Technol.* 109 (2013) 90–95.
- [50] G. Elordi, M. Olazar, G. Lopez, P. Castaño, J. Bilbao, Role of pore structure in the deactivation of zeolites (HZSM-5, H β and HY) by coke in the pyrolysis of polyethylene in a conical spouted bed reactor, *Appl. Catal. B: Environ.* 102 (2011) 224–231.
- [51] M. Abbas-Abadi, M. Kusenberg, A. Zayoud, M. Roosen, F. Vermeire, S. Madanikashani, M. Kuzmanovic, B. Parvizi, U. Kresovic, S. De Meester, Thermal pyrolysis of waste versus virgin polyolefin feedstocks: The role of pressure, temperature and waste composition, *Waste Manage.* 165 (2023) 108–118.
- [52] M. Kusenberg, A. Zayoud, M. Roosen, H. Thi, M. Abbas-Abadi, A. Eschenbacher, U. Kresovic, S. De Meester, K. Van Geem, A comprehensive experimental investigation of plastic waste pyrolysis oil quality and its dependence on the plastic waste composition, *Fuel Process. Technol.* 227 (2022) 107090.